



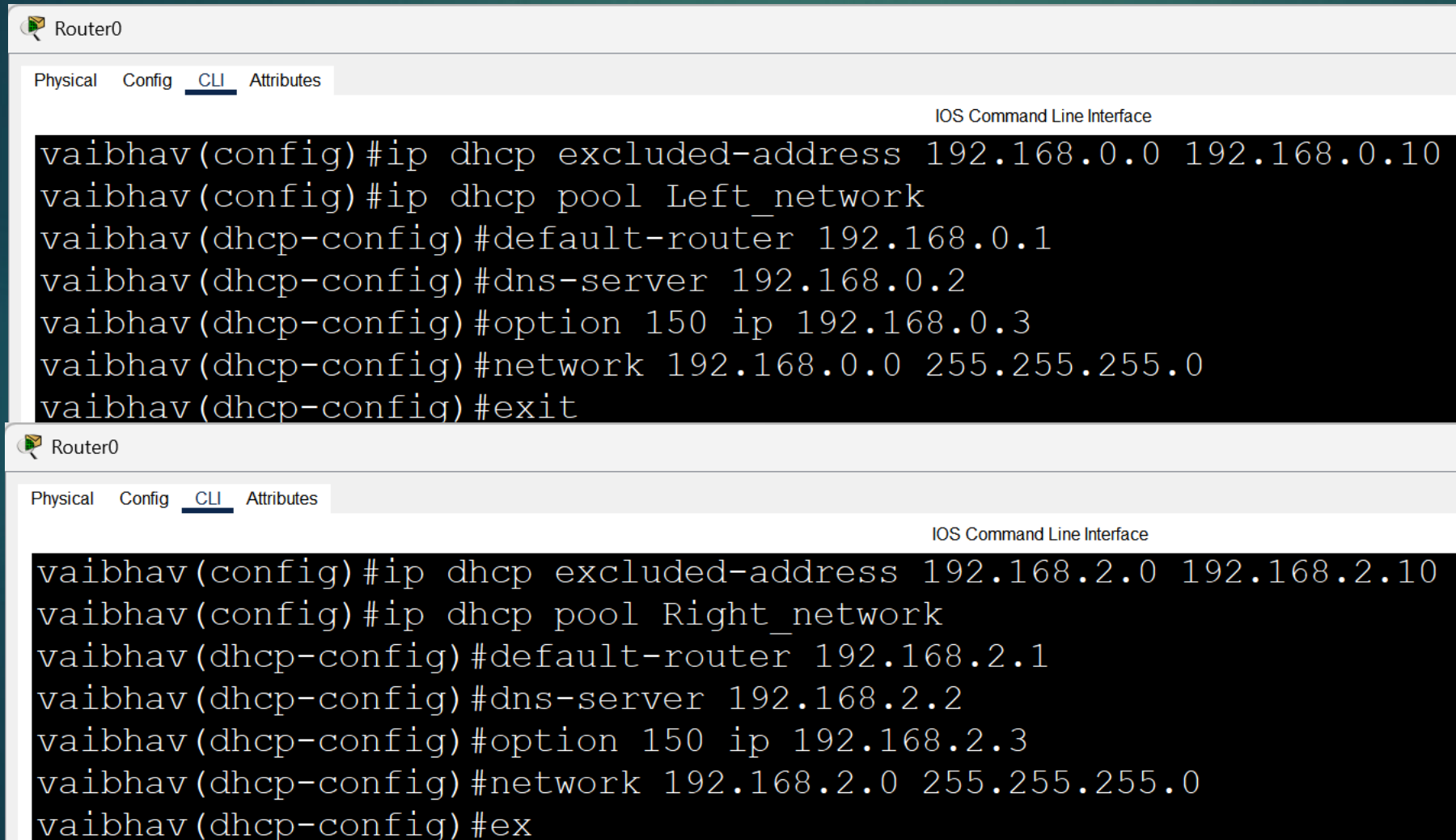
COMPANY NETWORK IMPLEMENTING ROUTING PROTOCOL AND ACCESS NETWORKS

VAIBHAV RAJENDRA SALUNKHE

Setting up the network

- ▶ Use Router
- ▶ Use Switch
- ▶ Use Hubs
- ▶ Use PC

Use DHCP Protocol and give IP address automatically



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
vaibhav(config)#ip dhcp excluded-address 192.168.0.0 192.168.0.10
vaibhav(config)#ip dhcp pool Left_network
vaibhav(dhcp-config)#default-router 192.168.0.1
vaibhav(dhcp-config)#dns-server 192.168.0.2
vaibhav(dhcp-config)#option 150 ip 192.168.0.3
vaibhav(dhcp-config)#network 192.168.0.0 255.255.255.0
vaibhav(dhcp-config)#exit

Router0
Physical Config CLI Attributes
IOS Command Line Interface
vaibhav(config)#ip dhcp excluded-address 192.168.2.0 192.168.2.10
vaibhav(config)#ip dhcp pool Right_network
vaibhav(dhcp-config)#default-router 192.168.2.1
vaibhav(dhcp-config)#dns-server 192.168.2.2
vaibhav(dhcp-config)#option 150 ip 192.168.2.3
vaibhav(dhcp-config)#network 192.168.2.0 255.255.255.0
vaibhav(dhcp-config)#ex
```

Dynamic
Host
Configuratio
n
Protocol

Use DHCP Protocol and give IP address automatically

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.0.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.0.1

DNS Server 192.168.0.2

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::20C:CFFF:FE4A:2C96

Default Gateway

DNS Server

802.1X

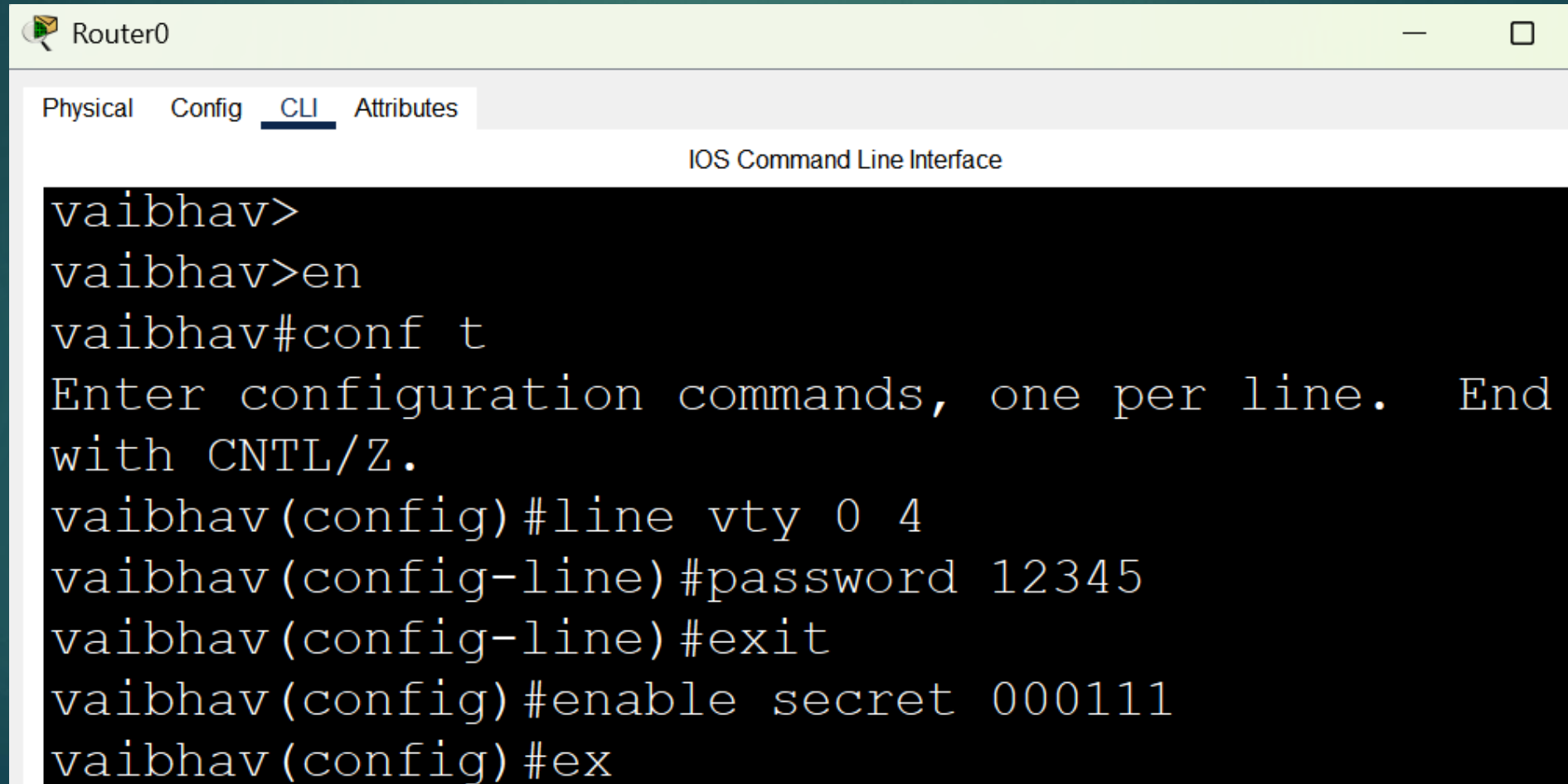
☐ Use 802.1X Security

Authentication MD5

Username

Password

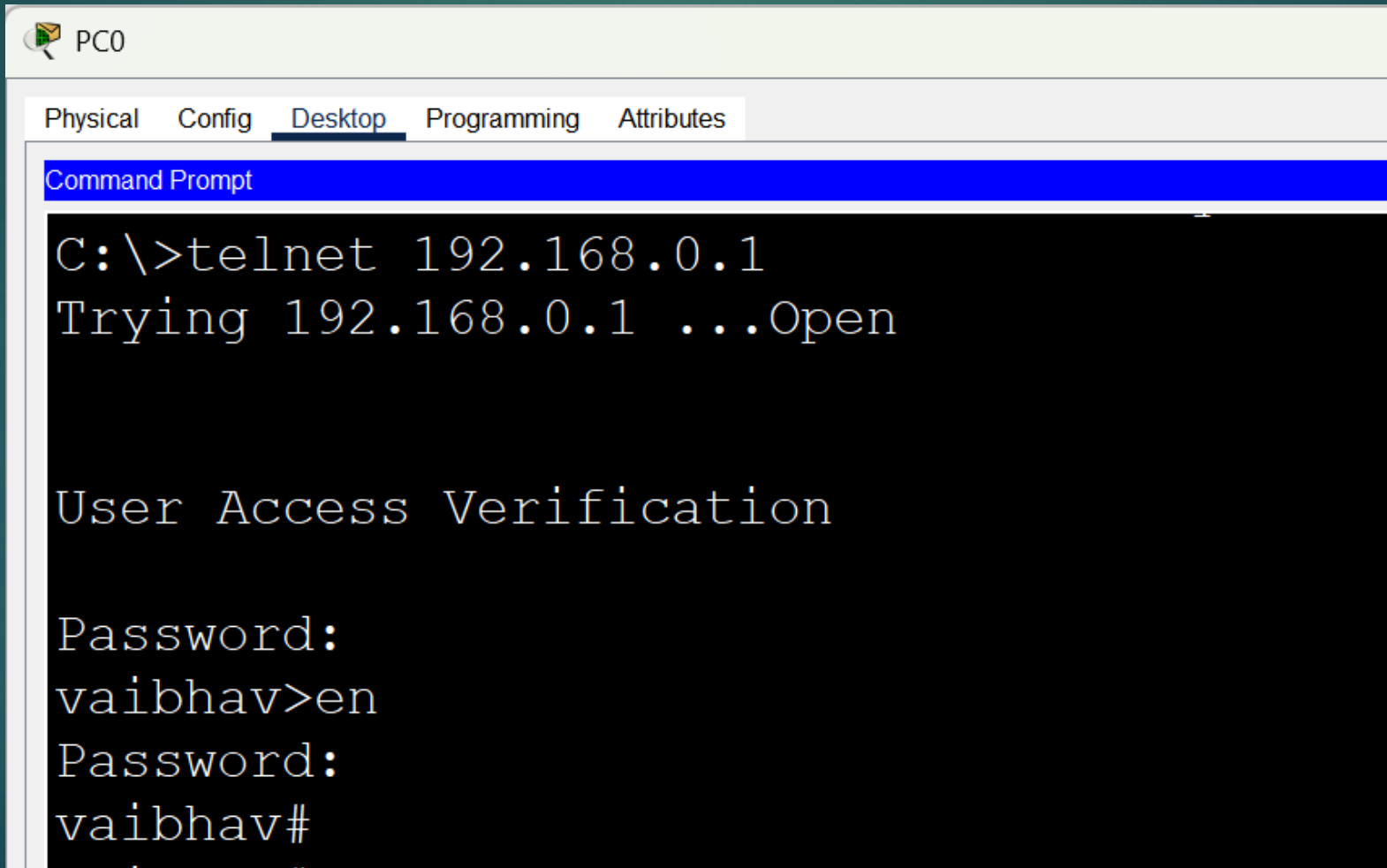
Telnet : add telnet commands in router



The screenshot shows a web-based interface for a Cisco router named 'Router0'. The 'CLI' tab is selected under the 'Config' section. The terminal window displays the following commands and prompts:

```
vaibhav>
vaibhav>en
vaibhav#conf t
Enter configuration commands, one per line.  End
with CNTL/Z.
vaibhav(config)#line vty 0 4
vaibhav(config-line)#password 12345
vaibhav(config-line)#exit
vaibhav(config)#enable secret 000111
vaibhav(config)#ex
```

Telnet: Access the router using telnet

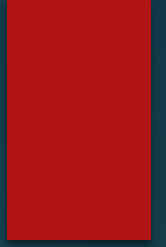


The screenshot shows a PC0 window with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt. The Command Prompt shows the execution of the telnet command to 192.168.0.1, followed by a successful connection to a router. The router prompts for a password, which is entered as 'vaibhav', and then shows the router's prompt as 'vaibhav#'. The text '...Open' is also visible on the line following the connection attempt.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
C:\>telnet 192.168.0.1
Trying 192.168.0.1 ...Open

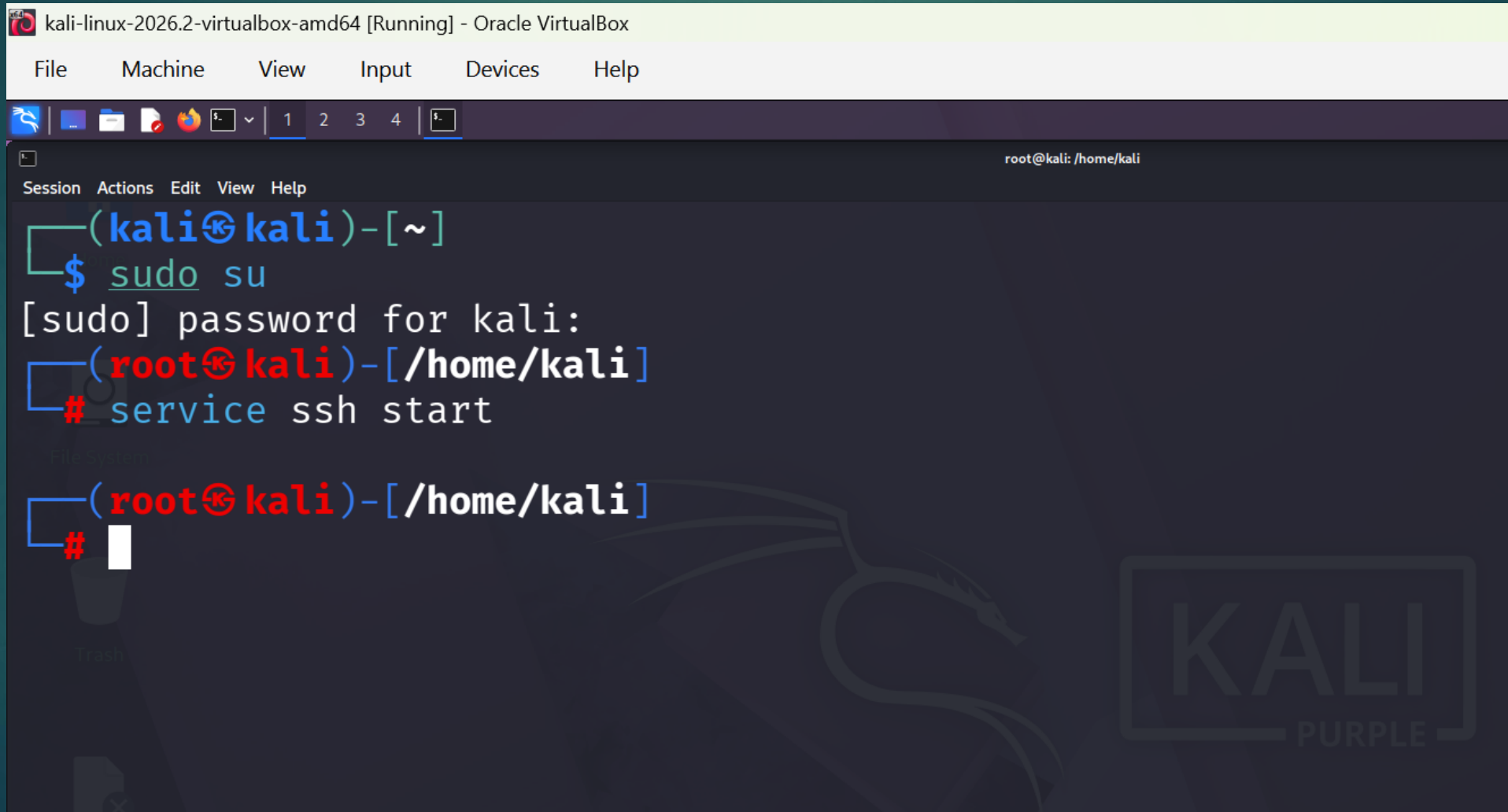
User Access Verification

Password:
vaibhav>en
Password:
vaibhav#
```



We can access
Linux from
Windows using
SSH commands.

Kali Linux : Open a terminal give ssh command



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# service ssh start

(root@kali)-[/home/kali]
#
```

Windows: Access Kali Linux using SSH

root@kali: /home/kali/Desktop

Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Vaibhav>ssh kali@192.168.0.101

kali@192.168.0.101's password:

Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.

Last login: Fri Sep 25 00:10:19 2026 from 192.168.0.143

(kaliⓈkali)~

\$ sudo su

[sudo] password for kali:

(rootⓈkali)-[/home/kali]

cd Desktop

(rootⓈkali)-[/home/kali/Desktop]

|

Access Kali Linux and use some commands

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4



Home



File System



Trash



confidential.txt



script.sh



vaibhav.txt



vaibhav

root@kali: /home/kali/Desktop

```
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\Vaibhav>ssh kali@192.168.0.101
```

```
kali@192.168.0.101's password:
```

```
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64
```

```
The programs included with the Kali GNU/Linux system are free software;
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```

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```

```
Last login: Fri Sep 25 00:10:19 2026 from 192.168.0.143
```

```
(kali㉿kali)-[~]
```

```
$ sudo su
```

```
[sudo] password for kali:
```

```
(root㉿kali)-[/home/kali]
```

```
# cd Desktop
```

```
(root㉿kali)-[/home/kali/Desktop]
```

```
# mkdir vaibhav
```

```
(root㉿kali)-[/home/kali/Desktop]
```

```
# touch vaibhav.txt
```

```
(root㉿kali)-[/home/kali/Desktop]
```

```
#
```

Access Kali Linux and use some commands

kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4



```
root@kali: /home/kali/Desktop x + v
[~](root@kali)-[/home/kali/Desktop]
# touch vaibhav.txt

[~](root@kali)-[/home/kali/Desktop]
# ls
confidential.txt  script.sh  vaibhav  vaibhav.txt

[~](root@kali)-[/home/kali/Desktop]
# tree
.
├── confidential.txt
├── script.sh
├── vaibhav
└── vaibhav.txt

2 directories, 3 files

[~](root@kali)-[/home/kali/Desktop]
# cat vaibhav txt
cat: vaibhav: Is a directory
cat: txt: No such file or directory

[~](root@kali)-[/home/kali/Desktop]
# rmdir vaibhav

[~](root@kali)-[/home/kali/Desktop]
# rm vaibhav.txt

[~](root@kali)-[/home/kali/Desktop]
#
```



▶ Thank you